

Tier	Function / Component	Manufacturer	Manufacturer Part Number	Description / Specification	Quantity	Unit Cost	Total Cost	Interface
2	978 MHz UAT antenna	Dynon	102607-000	Aircraft 978 MHz UAT ADS-B antenna	1	44.7	44.7	RF coax
2	978 MHz RF band-pass filter	Mini-Circuits	CBP-1034C+	Surface-mount RF band-pass filter for the UAT receive chain	1	48.7	48.7	50-ohm RF
2	RF low-noise amplifier	Analog Devices	HMC8411TCPZ-EP-PT	10 MHz to 10 GHz low-noise amplifier; Enhanced Product version	1	47	47	6-LFCSP; RF input/output
2	RF mixer / downconverter	Analog Devices	LTC5567	300 MHz to 4 GHz active downconverting mixer	1	9.28	9.28	RF / LO / IF
2	Local oscillator synthesizer	Analog Devices	ADF4351BCPZ-RL7	Wideband PLL with integrated VCO	1	25.16	25.16	SPI; RF LO output
2	Intermediate-frequency ADC	Analog Devices	AD9235BCPZ-20	12-bit 20 MSPS analog-to-digital converter	1	25.35	25.35	Parallel digital output
2	Digital demodulation FPGA	Microchip	A3P250-VQ100MIC	ProASIC3 250k-gate FPGA; military-temperature version	1	204.92	204.92	10-TQFP; parallel ADC input; digital I/O
2	RTOS control MCU	Texas Instruments	S5LS20216ASPGEMEP	TMS570LS20216-EP Hercules MCU	1	55.49	55.49	144-LQFP; SPI; UART; GPIO
2	ARINC 429 transmitter	Holt Integrated Circuits	HI-35870PQTF	Single ARINC 429 transmitter IC	1			SPI to MCU; ARINC 429 output
2	Ethernet MAC / PHY	Microchip	LAN9250I/ML	10/100 Ethernet controller with integrated PHY	1	8.17	8.17	SPI/SQI to MCU; Ethernet
3	ADF4351 reference TCXO	TBD	TBD	Low-phase-noise reference oscillator for RF synthesizer	1			Clock / RF
3	PLL loop-filter network	TBD	TBD	Calculated resistors and capacitors for ADF4351 PLL loop	1			Analog / RF
3	ADC sampling clock	TBD	TBD	Low-jitter sampling clock source for AD9235	1			Clock
3	IF filter / anti-alias network	TBD	TBD	Intermediate-frequency filtering before the ADC	1			Analog / RF
3	RF matching / bias network	TBD	TBD	Matching, biasing, coupling, and RF decoupling components	1			RF
3	Ethernet magnetics / connector	TBD	TBD	Ethernet isolation magnetics and external connector	1			Ethernet
3	Local power regulators	TBD	TBD	Required FPGA, MCU, RF, ADC, and ARINC voltage rails	1			Power
3	RF / ESD protection	TBD	TBD	Input surge and ESD protection for RF and external interfaces	1			RF / protection